

Morphic Studio Software Architecture Specification

Document 49 — Workflow Engine & Automation Architecture

Version: 1.0

Status: Approved

1. Purpose

This document defines the Workflow Engine and Automation Architecture for Morphic Studio.

Its purpose is to enable creators, teams, plugins, and AI services to automate creative processes using configurable, event-driven workflows that are reliable, auditable, and extensible.

The workflow engine serves as the orchestration layer connecting every major subsystem of the platform.

2. Workflow Philosophy

The workflow engine follows these principles.

2.1 Event-Driven Automation

Workflows respond to events generated throughout the platform.

Examples include:

- Project Created
- Character Updated
- Asset Uploaded
- AI Generation Completed
- Export Finished
- Publication Scheduled

Events initiate automation without requiring manual intervention.

2.2 Human-Centered Automation

Automation assists creators without removing creative control.

Workflows may pause for review or approval before continuing.

2.3 Visual First

Users create workflows through a visual workflow builder.

Advanced users may also define workflows using structured configuration.

2.4 AI-Assisted Automation

AI can recommend, generate, optimize, and explain workflows.

Users remain responsible for approving workflow changes.

2.5 Reliability

Workflow execution must survive application restarts, deployments, and temporary service interruptions.

3. Workflow Components

Each workflow consists of:

- Triggers
- Conditions
- Actions
- Variables
- Branches
- Loops (where appropriate)
- Approvals
- Notifications
- Error Handlers

Each component is independently versioned.

4. Trigger System

Supported triggers include:

Project Events

Story Events

Character Events

Asset Events

Publishing Events

AI Events

Time-Based Schedules

Webhook Events

Plugin Events

Manual Execution

Future external integrations may contribute additional triggers.

5. Conditions

Workflows evaluate conditions before executing actions.

Examples:

User role

Project status

Asset type

Language

Subscription tier

Workflow variables

AI confidence score

Date and time

Conditions should support logical composition.

6. Actions

Supported actions include:

Generate AI Content

Update Story Bible

Create Assets

Send Notifications

Publish Content

Export Project

Invoke Plugin

Call External API

Create Tasks

Schedule Jobs

Modify Metadata

Actions may be chained together.

7. Variables & Context

Workflow execution includes contextual variables such as:

Project

User

Characters

Assets

Timeline

Story Bible

Creative Memory

Workflow State

Previous Outputs

Variables may be referenced by later steps.

8. Workflow State Management

Execution state should support:

Running

Paused

Waiting

Approved

Rejected

Retrying

Completed

Cancelled

Failed

State transitions must be recorded.

9. Event-Sourced Execution

Every workflow execution generates immutable events.

Examples:

Workflow Started

Step Executed

Approval Requested

Approval Granted

Retry Performed

Failure Detected

Workflow Completed

These events create a complete execution history.

10. Scheduling

Support:

Immediate execution

Delayed execution

Recurring schedules

Cron expressions

Calendar schedules

Dependency-based scheduling

Scheduled workflows survive system restarts.

11. Human Approval

Certain workflow stages may require human review.

Approval steps support:

Reviewer assignment

Comments

Deadlines

Escalation

Approval history

Rejected workflows may return to previous stages.

12. Error Handling

Workflows define recovery behavior.

Options include:

Retry

Skip Step

Rollback

Pause

Escalate

Notify

Abort

Recovery policies are configurable.

13. AI Integration

The workflow engine integrates with the Unified AI Gateway.

AI capabilities include:

Workflow generation

Step optimization

Prompt construction

Content generation

Quality evaluation

Execution recommendations

AI activity remains observable and auditable.

14. Plugin Integration

Plugins may register:

Triggers

Actions

Conditions

Variables

Validators

Workflow Templates

All integrations use the public extension framework.

15. Monitoring & Analytics

Track:

Execution duration

Success rate

Failure rate

Retry frequency

Approval latency

Resource usage

AI costs

Workflow popularity

Metrics support optimization and capacity planning.

16. Versioning

Every workflow maintains version history.

Support:

Draft

Published

Archived

Rollback

Comparison

Migration

Existing executions continue using the version with which they started.

17. Acceptance Criteria

The workflow architecture is successful if:

- Workflows automate complex creative processes.
 - Execution survives failures and restarts.
 - Human approvals integrate seamlessly.
 - Plugins extend workflow capabilities.
 - AI assists without reducing user control.
 - Execution history is complete and auditable.
 - Workflows remain reusable and versioned.
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Conclusion

The Morphic Studio Workflow Engine establishes a flexible, event-driven automation framework that coordinates creative processes across the entire platform. By combining visual workflow design, event sourcing, AI assistance, plugin extensibility, and robust execution management, the platform enables creators to automate sophisticated production pipelines while maintaining transparency, reliability, and creative oversight.